$$P_n(5) = 288\tau(n), \quad G_n(U) = 5^{-\mu_n} P_n(5U)$$

$p = 499$

$\overline{G}(U) = 2 + 3U^2$

Simple root at 1

$p = 16249$

$\overline{G}(U) = (U-1)^2(U-2)$

Double root at 1

Auxiliary variable U. The target coefficient is evaluated at U = 1.